STAGE 1: Dart Fundamentals

Flutter & Dart: From Zero to Expert — Stage Textbook 1 of 8

Goal: Be comfortable writing basic Dart programs before touching Flutter.
Estimated time: 2–3 weeks

1.1 What is Dart?

Dart is a strongly-typed, object-oriented language made by Google. Flutter uses Dart exclusively. It compiles to native ARM code for mobile and to JavaScript for web.

Install Dart: https://dart.dev/get-dart Or use the online playground: https://dartpad.dev

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1.2 Variables & Types

```
void main() {
  // Explicit types
  int age = 25;
  double height = 5.9;
  String name = 'Alice';
  bool isStudent = true;

  // Type inference
  var city = 'Lagos';       // inferred as String
  final country = 'Nigeria'; // cannot be reassigned
  const pi = 3.14159;       // compile-time constant

  print('$name is $age years old and lives in $city');
}
```

Key rules:

- `var` — inferred, reassignable
- `final` — set once at runtime
- `const` — set at compile time, deeply immutable

1.3 Null Safety

Dart is null-safe by default. A variable cannot be null unless you explicitly allow it.

```
String name = 'Alice';   // cannot be null
String? nickname;        // can be null (nullable)

// Null-aware operators
print(nickname ?? 'No nickname');   // default if null
print(nickname?.length);            // safe access, returns null if nickname is null
nickname ??= 'Ali';                 // assign only if currently null
```

1.4 Control Flow

```dart
void main() {
  int score = 72;

  // if/else
  if (score >= 70) {
    print('Pass');
  } else {
    print('Fail');
  }

  // switch
  String grade = score >= 70 ? 'B' : 'C';
  switch (grade) {
    case 'A': print('Excellent'); break;
    case 'B': print('Good'); break;
    default:  print('Keep going');
  }

  // loops
  for (int i = 1; i <= 5; i++) {
    print(i);
  }

  int x = 0;
  while (x < 3) {
    print(x++);
  }
}
```

1.5 Functions

```dart
// Basic function
int add(int a, int b) => a + b;

// Optional named parameters
void greet({required String name, String greeting = 'Hello'}) {
  print('$greeting, $name!');
}

// Arrow functions (single expression)
bool isEven(int n) => n % 2 == 0;

void main() {
  print(add(3, 4));           // 7
  greet(name: 'Bob');         // Hello, Bob!
  greet(name: 'Sam', greeting: 'Hi'); // Hi, Sam!
  print(isEven(6));           // true
}
```

1.6 Hands-On Project: Temperature Converter CLI

Build a command-line Dart program that:

1. Defines a function `celsiusToFahrenheit(double c)` and `fahrenheitToCelsius(double f)`.
2. Asks nothing from the user (hardcode a list of test temperatures) and prints a neat table of conversions using a `for` loop.
3. Uses `final` for the list of temperatures and `const` for any fixed conversion ratio you pull out as a named constant.
4. Uses a `switch` statement to label each converted temperature as `"Freezing"`, `"Cold"`, `"Mild"`, or `"Hot"`.

This project forces you to combine variables, functions, control flow, and string interpolation — the whole of Stage 1 — in one small program.

1.7 Stage 1 Test

Answer all questions. Check answers by running code in DartPad.

Question 1: What is the difference between `final` and `const`? Give an example of each.

Question 2: What does the `??` operator do? Write a code snippet using it.

Question 3: Write a function `fizzbuzz(int n)` that prints:

- "Fizz" if n is divisible by 3
- "Buzz" if divisible by 5
- "FizzBuzz" if divisible by both
- The number itself otherwise

Question 4: What is null safety? What symbol makes a variable nullable?

Question 5: Write a `for` loop that prints the squares of numbers 1 to 10.

Passing score: Get all 5 correct before moving to Stage 2.